Playwright QA Automation for Insurance Order Entry Platform

Tech Forum / Course Project Listing Specification

Attribute	Project Specification Details
Course Title	Playwright QA Automation for Insurance Order Entry Platform
Domain & Category	Engineering / Fintech / Social Impact
Primary Skill Tag	Playwright, QA Automation, Java, Page Object Model
Level & Format	Medium
Sponsor / Enterprise Partner	MassMutual India
Pricing / Access	Free

Executive Summary

An insurance company operates a digital order entry platform that enables agents to capture, validate, and submit Whole Life insurance policy applications. Agents use the platform to log in, enter insured client details, capture address and KYC information, select insurance products, configure policy coverage, add riders, define beneficiaries, upload documents, validate the application, and submit the policy application for processing.
The company releases changes frequently across authentication, client information capture, product configuration, rider rules, beneficiary allocation, document upload, validation,

submission, and application status tracking. Because these are business-critical and compliance-sensitive journeys, manual regression testing is no longer sufficient.
The QA team wants a repeatable Playwright-based automation framework that can validate UI journeys, API responses, negative scenarios, regression scenarios, and release readiness.

Business Context & System Scope

The leadership team and product team expect the application to work reliably because agents depend on it for accurate policy submission. A defect in client data capture, product selection, rider eligibility, beneficiary allocation, KYC upload, or submission workflow can directly affect:

- **Application Quality & Accuracy:** Preventing incorrect coverage or demographic errors before submission.
- **Compliance Review:** Ensuring strict regulatory adherence for KYC and document verification.
- **Turnaround Time:** Minimizing processing delays caused by rejected or improperly validated policies.
- **Customer & Agent Trust:** Maintaining confidence and smooth platform operations for field agents.
- **Operational Workload:** Reducing manual QA cycles and administrative remediation work.

Core Learning Modules & Curric5ulum Breakdown

Learners will work through five key phases to construct a robust testing framework:

1. **Insurance Workflow & QA Planning:** Study end-to-end business workflows, evaluate compliance touchpoints, and construct a detailed QA test plan.
2. **Playwright UI Automation:** Automate multi-step user interface flows including login, client data entry, rider configuration, and document upload.
3. **API Testing & Page Object Model (POM):** Architect a scalable Page Object Model structure and execute backend API validations.
4. **Regression & Negative Testing:** Build edge-case test suites, validate system resilience, and establish automated regression coverage.
5. **CI/CD, Reporting & Release Readiness:** Integrate test runs into continuous integration pipelines, debug test failures, generate reporting dashboards, and evaluate release stability.

Expected Outcomes

This project will help learners apply Playwright automation skills in a realistic insurance product environment. Learners will study business workflows, prepare a QA plan, design test cases, automate UI and API validations, implement Page Object Model, run tests through CI/CD, debug failures, generate reports, and document defects with business impact.